PriorityKV: When Structure-Aware KV Retention Helps, and the Measurable Threshold Where It Stops

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Abstract

Long agent traces pack tool schemas, persistent identifiers, superseding instructions, and filler into one key–value (KV) cache. Every token has the same storage cost, but not the same failure cost. This motivates evicting by *application-visible structure*: protect schemas, constraints, sinks, and the recent window; discard the rest. We show that this works dramatically in one regime, fails completely in another, and that a single measurable quantity predicts which regime a workload is in.

On a synthetic stress benchmark (Qwen3-8B, matched 25% keep), structure-aware retention passes 112/120 examples against 1/120 for position-blind eviction, and is statistically indistinguishable from reimplementations of SnapKV, PyramidKV, and a structure–SnapKV hybrid at 108/120 (exact McNemar $p = 0.125$). On BFCL V3 multi-turn, scored by the unmodified official checker, the same policy scores **0.000** on both Qwen3-8B (n=140) and Llama-3.1-8B (n=143), while attention-based eviction remains indistinguishable from FullKV at the same budget ($p = 0.152$ and $p = 1.0$).

The cause is measurable before any generation. A role-based policy can rank only while protected content is a *minority* of context; above that it selects everything and degenerates to index order. Our synthetic benchmark is 94.9% filler (6.1% protected); a BFCL system prompt *is* 32 JSON tool schemas (98.8% protected). We propose this protected fraction as a CPU-computable pre-flight criterion, and ADAPT, a budget-relative prior that reduces to the structure policy when it can rank and to attention selection when it cannot; ADAPT ties SnapKV on BFCL ($p = 1.0$) where the pure structure policy scores zero.

We map the operating envelope with the same rigor. Structure matches FullKV under mid-context relocation and degrades when state is buried in free-form prose, bounding the prior to recognizable state. Role-aware INT4 placement is separately tested and shown not to be a quality lever, which keeps the storage and retention claims independent. The packed BF16/INT4 prototype reduces KV payload to 0.719× at 1.200–1.215× time per output token, an explicit space–time trade rather than an unqualified win. We also publish a correction to one released baseline (§9.1), so that every column in this paper can be audited against the code.

1 Introduction

Forty turns into an illustrative agent conversation, the model emits a tool call without a required field. The field belonged to a schema that the cache evicted thirty turns earlier to stay under budget; byte for byte, it looked no different from the filler that survived. This is the failure PriorityKV targets. In an agent trace, a JSON schema field and a throwaway pleasantry cost the same storage but not the same failure. Losing the former can also resurrect an obsolete instruction or substitute the wrong persistent identifier. Figure 1 shows this running example schematically; it is illustrative, not an additional measured trace.

Autoregressive inference stores prior keys and values so each decode step can reuse the prefix. The cache grows with sequence length and constrains serving capacity [1, 2]. Existing work reduces that cost through paging, streaming windows, attention-derived eviction, or quantization [3, 4, 5, 6, 7]. Agent traces add a signal those methods do not directly use: their serialization already exposes message roles, tool contracts, constraints, results, and state.

PriorityKV asks whether application-visible structure is useful when a cache must discard tokens, and separately whether a mixed BF16/INT4 implementation can save bytes. These are different hypotheses: eviction removes information, whereas quantization retains an approximation. Combining their outcomes into one "compression quality" claim would conceal the falsified INT4 hypothesis and the measured latency cost. Figure 2 summarizes the three experiment families and their distinct claim boundaries.

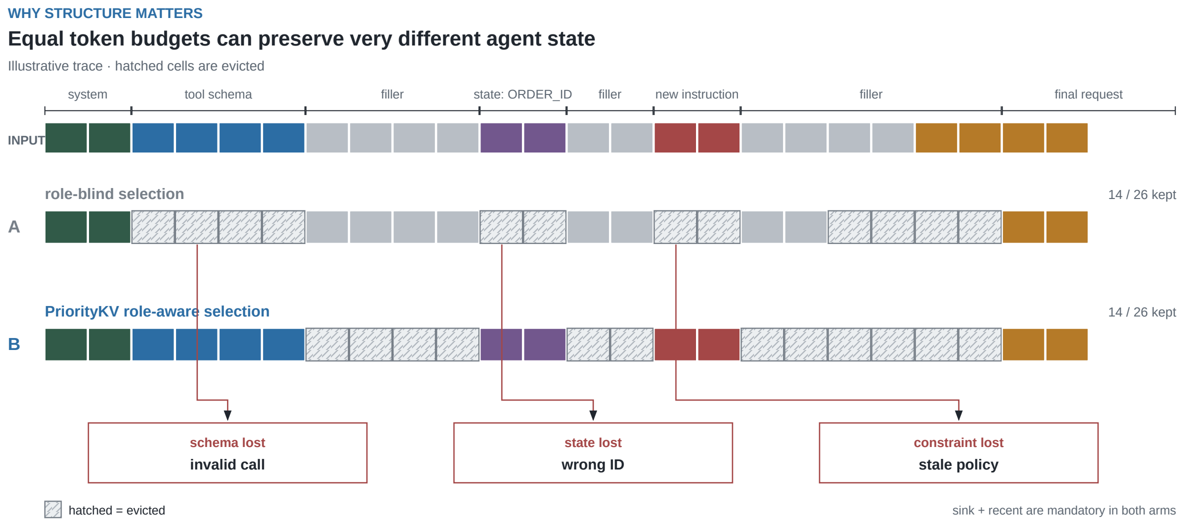

Figure 1: Equal storage cost, unequal failure cost. Two policies keep the same illustrative 14/26-token budget. Hatched cells are evicted; callouts show the resulting failure modes rather than empirical counts.

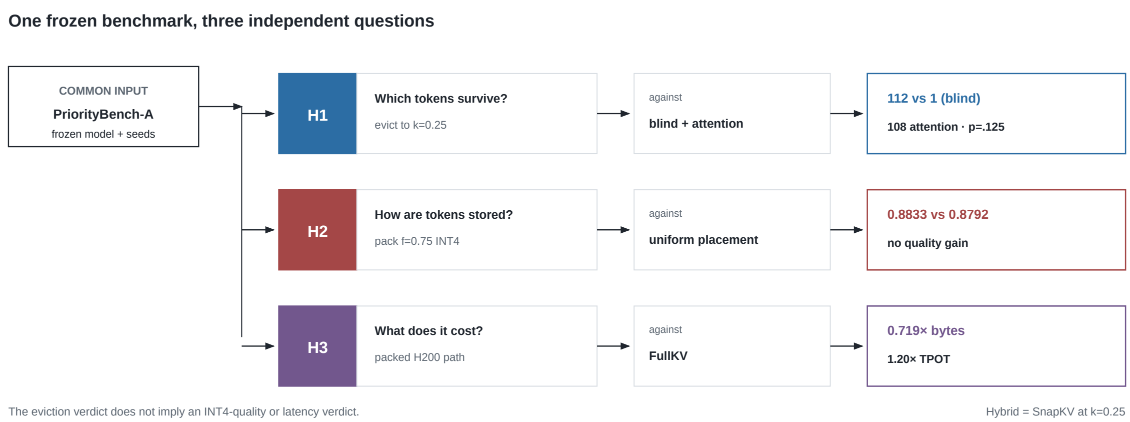

Figure 2: Three questions, three verdicts. Eviction reliability, precision placement, and systems performance are evaluated independently; displayed counts and ratios are measured endpoints.

Scope of claims. The paper separates three questions: whether structure helps eviction, whether role-aware precision placement preserves quality, and whether packed storage improves systems performance. A result in one family does not establish either of the others. Our baselines are matched-budget repository reimplementations, the benchmark is synthetic, and the systems measurements are single-request prototype results. We state additional limitations in Section 9 and do not repeat this boundary in every result.

This work makes four contributions.

1. It defines PriorityBench-A, a deterministic synthetic benchmark for tool-schema conformance, instruction supersession, and multi-turn state recall, together with disjoint stress slices, placement controls, and a gold-span retention audit.

2. It scales the decisive Qwen eviction comparison to $n = 120$ and adds matched-budget attention baselines plus a paired McNemar test. The supported claim is that structure matches or slightly exceeds SnapKV-class selection while decisively beating position-only baselines; it is not that structure beats FullKV or SnapKV in general.

3. It reports two important negatives: structure loses to SnapKV on both Llama slices at a 5% budget, and structure-aware placement of 75% INT4 positions does not create a quality advantage on the locked Qwen benchmark.

4. It implements packed BF16/INT4 pages and two-call FlashInfer decode, then reports payload, allocator peak, packing, cold-scratch, end-to-end, and per-output-token measurements separately.

2 Related Work

Paging and streaming. PagedAttention separates logical KV blocks from non-contiguous physical storage to reduce fragmentation and enable sharing; it does not choose semantic tokens to retain [2]. StreamingLLM identifies attention sinks and combines initial tokens with a recent window for streaming language modeling [3]. PriorityKV uses sink plus recent as a mandatory region and as the shape of its position-only control, not as a claim to reproduce StreamingLLM's streaming setting.

Eviction and attention-derived selection. H2O retains accumulated-attention heavy hitters together with recent tokens [4]. SnapKV uses an observation window near the end of the prompt to select clustered prefix features [5]. PyramidKV varies the cache budget across layers in response to pyramidal information funneling [6]. These methods infer importance from model behavior; PriorityKV tests application-visible message structure. The attention-baseline comparison uses repository reimplementations at matched token budgets, not the authors' reference code. In particular, H2O is a chunked SDPA reimplementation and carries an explicit fidelity caveat. A hybrid that protects structural tokens and fills the remainder with SnapKV was evaluated: it equals SnapKV on Qwen at 25% keep and collapses below both parents on the two Llama 5% slices. Thus the proposed complementarity hypothesis is not supported.

KV quantization and attention engines. KIVI motivates asymmetric group-wise low-bit KV quantization and keeps residual tokens at full precision [7]. PriorityKV's INT4 path is not a new quantizer and is not positioned as a replacement for KIVI. It is a real packed storage path for testing role-aware precision placement and for accounting payload bytes. FlashInfer exposes composable attention kernels and log-sum-exp state merging for heterogeneous KV layouts [8]. PriorityKV uses that merge contract, while retaining an explicit BF16 cold scratch limitation.

Long-context evaluation. LongBench and RULER test broad long-context capabilities [9, 10]. Our suite instead targets exact failure attribution in synthetic agent protocols. It cannot establish workload prevalence or replace general long-context suites.

3 PriorityBench-A

3.1 Tasks and scoring

PriorityBench-A generates multi-turn chats in which an early span contains the state required by a final request and intervening turns provide benign filler. Scoring is binary and programmatic; no LLM judge is used. Table 1 summarizes the three equally represented categories in the locked 240-example manifest.

Table 1: PriorityBench-A task families in the locked manifest ($n = 240$; 80 examples per category).

Category	Required behavior	Deterministic scorer
Tool schema	Emit a valid call under an early contract	JSON/schema subset
Instruction supersession	Follow the latest conflicting constraint	Required/forbidden regex
Multi-turn state	Reuse an earlier ID, path, or preference	Exact slot match

Two manifests from the same PriorityBench-A generator serve different studies. The locked 240-example manifest (W3) contains 83, 81, and 76 examples at nominal 8k, 16k, and 32k context strata; stable hashing assigns 92 calibration, 49 validation, and 99 test examples. The 120-example stress manifest (W5) is split evenly across the three categories and 8k/16k strata, with three disjoint 40-example slices. Reproducibility metadata, including both manifest digests, appears in Table 7 rather than in the narrative.

3.2 Placement and leakage controls

The generator's visible templates are also a threat: a heuristic tagger can appear semantic if gold state is consistently short, early, or marked by a template recognizable to the policy. We therefore evaluate two transformations of each stress slice. *Middle relocation* moves state away from the prefix while retaining the leading system message and final request. *Buried state* lengthens or embeds state-bearing turns so that a short-span heuristic does not protect all gold.

A separate CPU audit maps the gold-bearing messages into tokenizer spans and intersects them with the selected positions. This audit uses no model outputs. It measures the gold fraction already in sink plus recent and the gold fraction retained by each policy. Thus placement robustness and label leakage into the always-kept window are tested independently.

4 Method

4.1 Roles and matched-budget retention

Let a tokenized trace be $x_{1:n}$ with structural roles $r_{1:n}$. For keep fraction k, the nominal budget is

$$B(n,k) = \mathrm{round}(kn). \tag{1}$$

The mandatory set is $M = M_{\mathrm{sink}} \cup M_{\mathrm{recent}}$, where the first 16 tokens and last 128 tokens are protected. In the evaluated token-level regime, each compressed arm selects $A \supseteq M$ with

$$|A| = \max(B(n,k), |M|). \tag{2}$$

Uniform retention uses the mandatory sink and fills from the recent edge. Random retention keeps M and samples the remaining budget. Structure-aware retention first includes system, tool, constraint, and other state-bearing positions, then fills any residual budget from the recent edge. Whole-page experiments apply the same priority order while respecting page boundaries and never exceeding the token budget except for M.

The tagger consumes message roles and transparent schema or constraint markers. It does not inspect benchmark answers or future attention. The policy protects explicit protocol roles more reliably than unmarked free-form state; the buried controls in Section 6.1 quantify that boundary.

4.2 Pages and mixed precision

The physical page size is $P = 16$ tokens. In the mixed-precision study, every position remains present and an exact target fraction $f = 0.75$ is stored as INT4, subject to the common sink/recent BF16 protection. The role-blind arm spaces INT4 positions across the demotable region. The structure arm demotes filler, generated, and low-risk positions first, spilling into protected soft roles only if the byte budget requires it. Both arms therefore have the same number of INT4 positions.

Figure 3 traces the implemented path from tagging to the page table. It separates policy decisions—retention for eviction or dtype for mixed precision—from physical payload. Cold K and V pages contain packed 4-bit values plus per-group FP32 scale metadata; hot pages remain BF16.

4.3 Two-call FlashInfer decode

Prefill uses the native Hugging Face path. The resulting cache is partitioned into coalesced hot BF16 pages and packed cold pages. Before the frozen full-scratch decode measurement, cold pages are dequantized into BF16 GPU scratch. Each layer makes at most two homogeneous FlashInfer calls: one over hot pages plus the decode tail and one over cold scratch. Their outputs and native log-sum-exp states are combined with `merge_state`. A parity artifact records maximum absolute error of approximately 4.88×10^{-4} after correcting an LSE-base mismatch.

Figure 4 shows why cache payload, allocator peak, and latency must be reported separately. Packed bytes persist, full cold scratch is live during decode, and the decode tail grows token by token. The streamed-cold experiment (P2) instead dequantizes chunks and releases them after merging, but its three-example run is only a successful smoke test and is not a comparative systems result.

5 Experimental Design

All GPU experiments use greedy decoding on NVIDIA H200 hardware. Canonical jobs use at most two GPUs, while each eviction, attention-baseline, and transfer job uses one. Table 7 records the pinned model revisions, manifest digests, and result bundles.

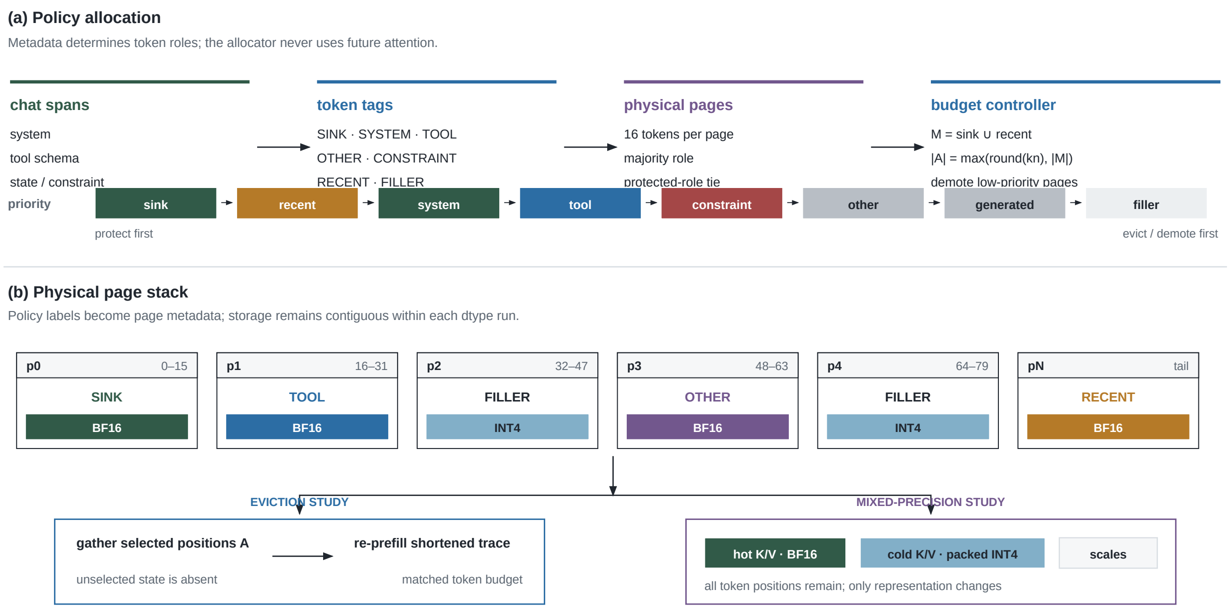

Figure 3: PriorityKV architecture. Chat metadata becomes token roles and 16-token pages before matched-budget retention or precision placement. The page records are illustrative.

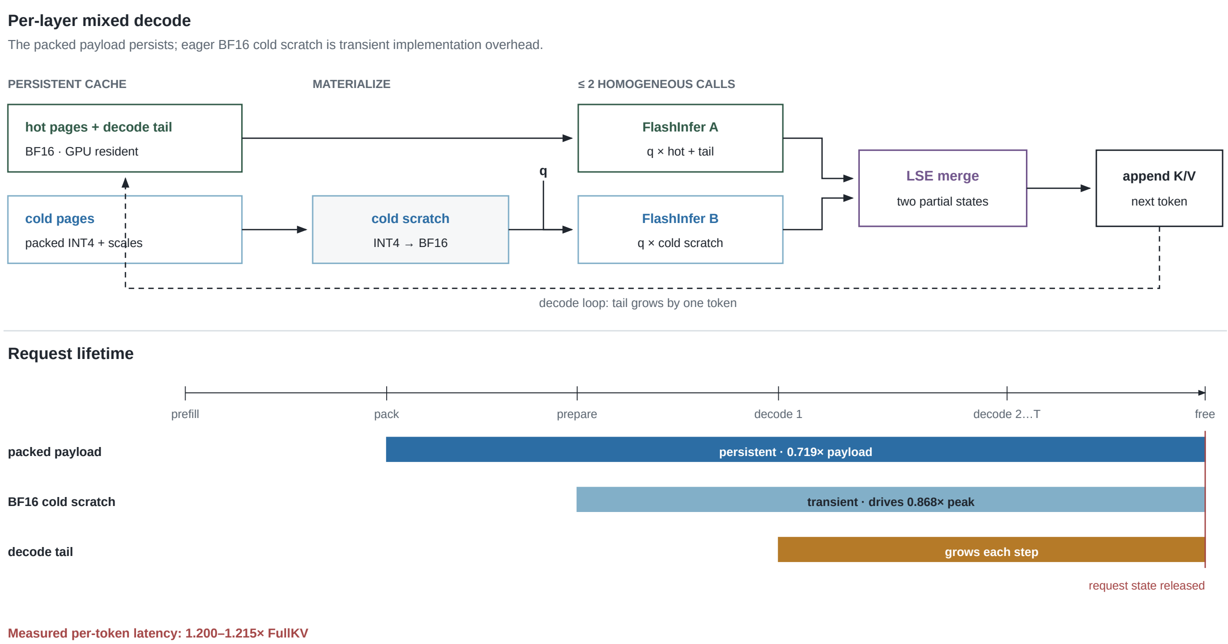

Figure 4: Decode path and allocation lifetime. Two homogeneous FlashInfer calls are merged through log-sum-exp state; packed payload persists while BF16 cold scratch is live during decode.

5.1 Eviction, attention baselines, and placement controls

The eviction and placement study (P0) evaluates Qwen on the 120-example stress manifest, divided into three disjoint 40-example slices and balanced across 8k and 16k contexts. At token keep $k = 0.25$, it compares structure, uniform, random, keep-all, and a separately recorded FullKV run using deterministic task pass rate. Middle-relocation and buried-state transformations are evaluated on all three slices. The attention-baseline comparison (P1) reuses the same examples, model, contexts, and budget for SnapKV, PyramidKV, H2O, and a structure–SnapKV hybrid. SnapKV uses a 64-token observation window and kernel size five. The H2O arm is a chunked SDPA reimplementation and carries a fidelity limitation. A tracked artifact supplies the exact two-sided paired McNemar test for structure versus SnapKV.

5.2 CPU retention audit

The audit runs the Qwen and Llama tokenizers on the three stress slices for each model ($n = 120$ each), maps gold messages to token spans, and applies uniform, random, and structure selection at $k = 0.25$. It reports mean token-retention fractions rather than model accuracy. These are descriptive span means, so no binomial interval or significance test is attached.

5.3 Mixed-precision quality and systems measurements

The locked mixed-precision study evaluates Qwen on all 240 examples: 83 at 8k, 81 at 16k, and 76 at 32k. FullKV, uniform-mixed, and structure-mixed arms use matched $f = 0.75$ INT4 placement for the mixed arms. The H200 packed FlashInfer path is scored by the same deterministic benchmark metric. This family tests a precision-placement quality hypothesis, not eviction. No paired significance artifact exists; we report Wilson intervals from the observed success counts.

The latency measurement uses nine 8k and nine 16k examples, 128 generated tokens, one warm-up, and three timing repetitions summarized by per-example medians. FullKV uses SDPA while the packed path uses the FlashInfer shim; this is an end-to-end prototype comparison, not a kernel-isolated study. The memory measurement uses 18 examples and records packed payload plus allocated and reserved CUDA peak. Both are single-request studies without concurrent serving or tail latency. Canonical jobs were not independently repeated, so no error bars are shown. A three-example streamed-cold run (P2) checks exit status and parity but serves only as a smoke test.

5.4 Cross-model transfer and weak checks

The Llama transfer study (P3) evaluates structure, SnapKV, H2O, PyramidKV, and hybrid on all three stress slices at $k = 0.25$ ($n = 120$). At $k = 0.05$, only the first two slices are run ($n = 40$ each); no uniform Llama arm is available. Two earlier checks remain deliberately secondary: one fixed 14-example Qwen slice is reused across three whole-page budgets, and a 14-example Gemma-2-9B-IT check uses Qwen-tuned prompts and has a low FullKV score.

6 Results

6.1 Structure protects state that blind eviction drops

At matched 25% token keep, structure passes 112/120 Qwen examples while the position-blind control passes 1/120 (Table 2). The corresponding Wilson intervals do not come close to overlapping. We report a single position-blind control rather than two: our released "random" arm is byte-identical to the uniform arm, a defect characterized in §9.1 rather than left for a reader to discover. Figure 5 places the position-blind and attention-derived selectors on one pass-rate axis.

The CPU audit confirms that this gap reflects retention rather than an easy always-kept window. Only 1.0–1.3% of Qwen gold tokens and none of the Llama gold tokens lie in sink plus recent (Table 3). Structure retains all mapped gold tokens, whereas uniform retains only the small fraction already in that window.

Placement controls bound the positive result (Table 4). Moving state to the middle gives both structure and FullKV 115/120, or 0.958 [0.906, 0.982]. Burying state in longer free-form turns lowers structure to 80/120, or 0.667 [0.578, 0.745], versus FullKV at 107/120, or 0.892 [0.823, 0.936]. Uniform scores 3/120 in the middle condition

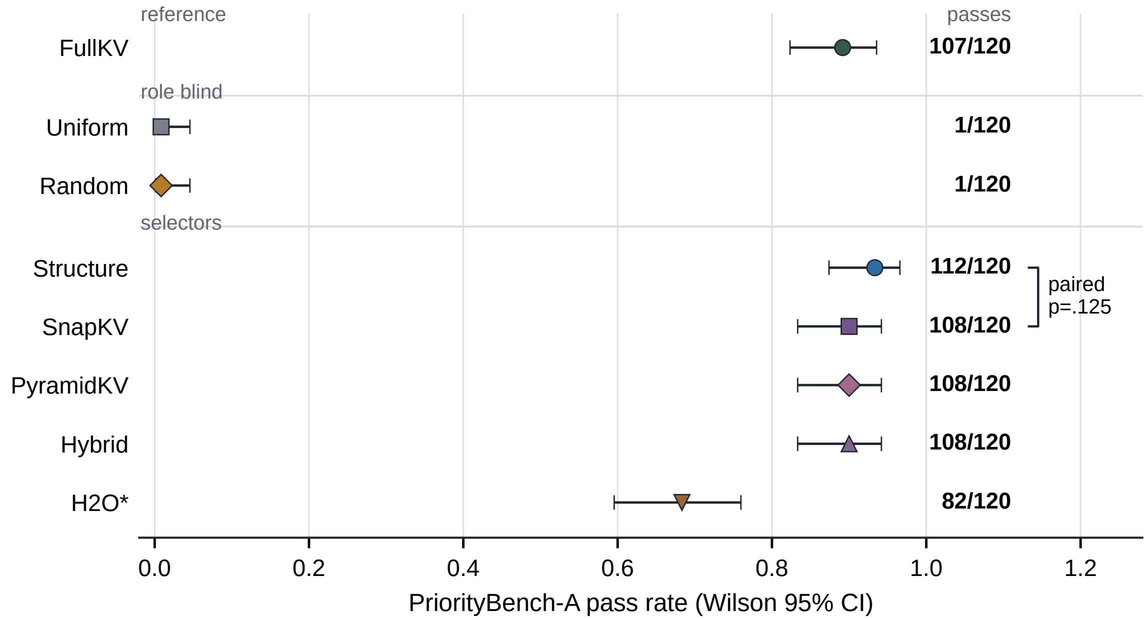

Figure 5: Matched-budget Qwen comparison ($n = 120$, $k = 0.25$). Points show pass rates with Wilson 95% intervals; exact counts are aligned at right and the bracket reports the paired structure–SnapKV McNemar test. H2O* denotes the chunked repository reimplementation.

Table 2: Pooled Qwen eviction and attention-baseline results at $k = 0.25$ ($n = 120$). Intervals are Wilson 95%.

Arm	Passes	Rate	Wilson 95% interval
FullKV	107/120	0.892	[0.823, 0.936]
Structure	112/120	0.933	[0.874, 0.966]
Position-blind†	1/120	0.008	[0.002, 0.046]
SnapKV	108/120	0.900	[0.833, 0.942]
PyramidKV	108/120	0.900	[0.833, 0.942]
Hybrid	108/120	0.900	[0.833, 0.942]
H2O (chunked)	82/120	0.683	[0.596, 0.760]

†Earlier versions of this table reported "uniform" and "random" as two arms, both at 1/120. They are a single computation: our `select_random` is byte-identical to `select_uniform` at every context length (§9.1). We report one row rather than double-count one control.

Table 3: Gold-span retention audit at $k = 0.25$ ($n = 40$ per slice). Entries are mean token fractions, not pass counts.

Model	Slice	Gold in sink+recent	Structure kept	Uniform kept
Qwen	0	0.010	1.000	0.010
Qwen	1	0.013	1.000	0.013
Qwen	2	0.010	1.000	0.010
Llama	0	0.000	1.000	0.000
Llama	1	0.000	1.000	0.000
Llama	2	0.000	1.000	0.000

and 0/120 when state is buried. Visible roles are therefore a useful prior, not an oracle for arbitrarily embedded state.

Table 4: Placement controls ($n = 40$ per slice), reported as structure / FullKV / uniform pass rates.

Slice	Middle relocation	Buried state
0	0.975 / 0.975 / 0.025	0.675 / 0.900 / 0.000
1	0.950 / 0.950 / 0.050	0.650 / 0.875 / 0.000
2	0.950 / 0.950 / 0.000	0.675 / 0.900 / 0.000

6.2 Structure matches attention selection on this benchmark; the hybrid does not help

Scope note. The parity reported here is specific to PriorityBench-A and does not survive transfer. §7 measures the same two policies on an externally authored benchmark and finds attention selection intact while the structure policy collapses to zero. The two results are consistent once the protected fraction is measured (§7.2); readers should carry the qualifier, not the headline.

Structure passes 112/120, while SnapKV, PyramidKV, and hybrid each pass 108/120 (Table 2). The structure interval overlaps every 0.900 arm. The paired structure–SnapKV artifact records $b = 4$ cases where structure alone passes and $c = 0$ in the opposite direction; exact two-sided McNemar $p = 0.125$. The difference is not statistically significant. Hybrid equals SnapKV rather than exceeding it, falsifying the complementarity hypothesis. H2O passes 82/120 using the chunked implementation; its implementation caveat prevents a reference-H2O claim.

6.3 The advantage depends on model and budget

All five tested structure and attention arms pass all 120 Llama examples at $k = 0.25$; the Wilson interval for 120/120 is [0.969, 1.000]. With no uniform arm, this is an easy-task ceiling rather than evidence of universal transfer. At $k = 0.05$, SnapKV passes 40/40 on each evaluated slice, while structure passes 35/40 and 36/40. No paired test was produced for these Llama slices.

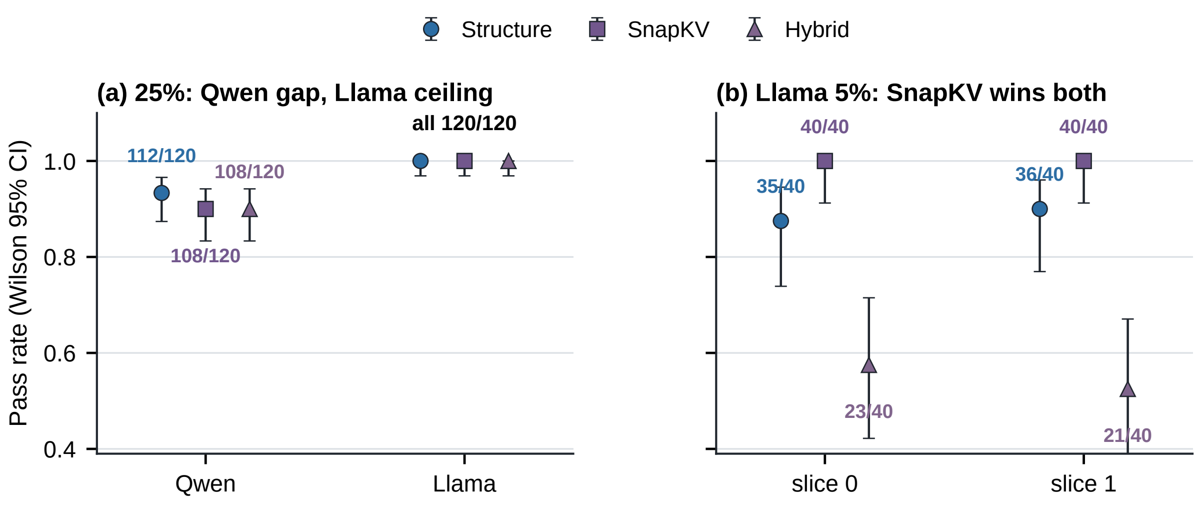

Figure 6: Budget and model dependence. Qwen and Llama at $k = 0.25$ use $n = 120$; each Llama $k = 0.05$ condition uses $n = 40$. Error bars are Wilson 95% intervals.

At the tighter budget, failures concentrate in tool-schema tasks. Structure passes 8/13 and 9/13 schema examples in the first and second slices, while SnapKV passes 13/13 in both. Hybrid is worse at 1/13 and 0/13, with overall rates 0.575 and 0.525. Protecting broad structural spans can therefore consume too much residual budget, and simply unioning role and attention selections does not make them complementary.

6.4 Matched INT4 placement does not separate quality

On the locked 240 examples, FullKV passes 213, uniform-mixed 211, and structure-mixed 212 (Table 5). The structure–uniform difference is one example and no paired statistical artifact supports a quality distinction. At 8k and 16k, every arm is 1.0; at 32k, FullKV, uniform, and structure are 49/76, 47/76, and 48/76. Figure 7 shows the count-backed Wilson intervals by context. The tested role-aware INT4 quality hypothesis is falsified.

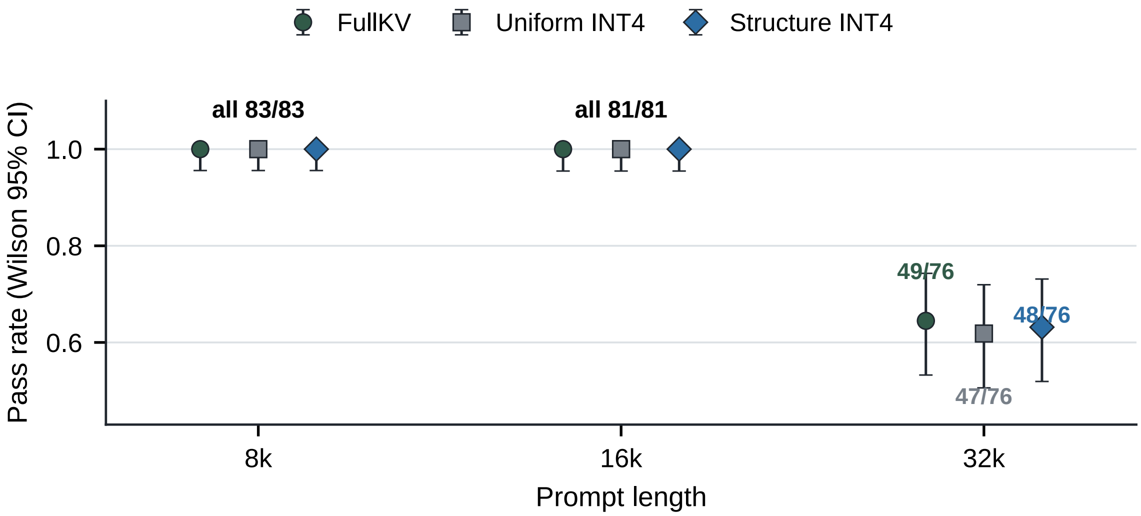

Figure 7: Matched INT4 quality. Qwen3-8B with FullKV or matched $f = 0.75$ INT4 placement. Error bars are Wilson 95% intervals from exact counts (83, 81, and 76 examples).

Table 5: Locked Qwen quality ($n = 240$). Intervals are Wilson 95%.

Arm	Passes	Rate	Wilson 95% interval
FullKV	213/240	0.8875	[0.841, 0.922]
Uniform mixed	211/240	0.8792	[0.832, 0.915]
Structure mixed	212/240	0.8833	[0.837, 0.918]

6.5 Packed bytes incur scratch and latency costs

The mixed structure path has a modeled ratio of 0.473×, measured packed payload ratio of 0.719×, and allocated-peak ratio of 0.868× FullKV on the 18-example memory slice. The gap is caused by BF16 cold scratch, not by an accounting inconsistency. Figure 8 reports memory and latency ratios on the same parity reference. No uncertainty bars are shown because the canonical jobs were not independently repeated.

At 8k, structure packing and scratch construction take 34.8 and 14.2 ms. E2E TTFT is 1.118× FullKV, and TPOT rises from 25.25 to 30.29 ms (1.200×). At 16k, pack and scratch take 48.1 and 19.7 ms; E2E is 1.113×, and TPOT rises from 23.84 to 28.95 ms (1.215×). These are regressions, not speedups. The streamed-cold smoke run exits successfully with parity and approximately 36.4 GiB peak in its log, but with only three examples and no comparative frontier it remains a smoke test rather than a systems result.

6.6 Additional weak checks

A whole-page Qwen ablation reuses the same 14 examples at 15%, 25%, and 35% keep. Structure remains 9/14 at every budget; random rises from 1/14 to 6/14, and the role-blind arm remains 0/14. Because the slice is reused, this is one small ablation rather than three replications. On a separate 14-example Gemma check, FullKV passes 5/14, structure 2/14, and uniform 0/14. The low FullKV score, Qwen-tuned prompts, and sample size make both results directional checks, not transfer evidence.

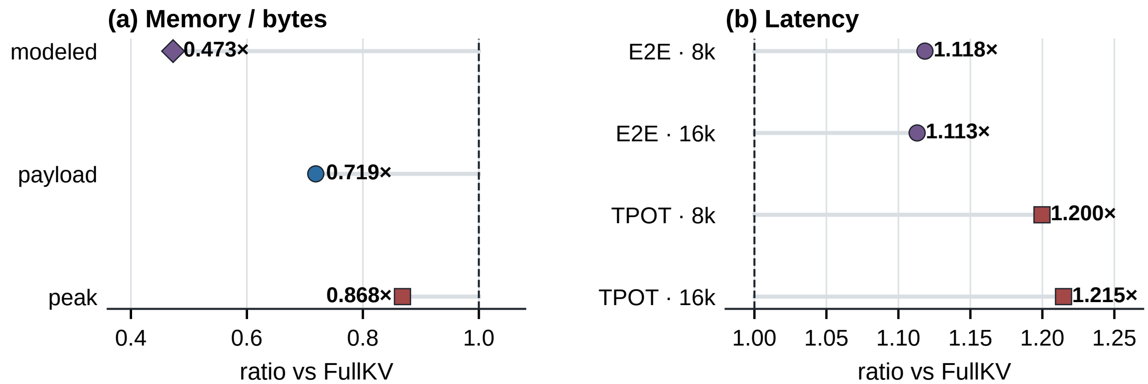

Figure 8: H200 systems trade-off. Packed payload is smaller than FullKV, allocator peak falls less, and every measured latency ratio exceeds the dashed 1.0 reference.

7 External Evaluation

The results above are measured on PriorityBench-A, a benchmark we constructed. This section tests whether the eviction prior transfers to a benchmark we did not author. We evaluate on BFCL V3 multi-turn using the unmodified official `multi_turn_checker`, which executes tool calls against stateful API implementations and compares final state and execution responses against ground truth. All non-FullKV arms are `kvpress` presses over an identical full prefill at an identical compression ratio, so the arms differ only in which KV entries survive, never in how they are removed.

7.1 The structural signal is exhausted in schema-dominated contexts

Table 6 reports 150 conversations per model at a 25% keep budget, scored per conversation.

Table 6: BFCL V3 multi-turn, official scorer, 25% keep. Paired completeness 0.933 (Qwen) and 0.953 (Llama); exclusions are all `MODEL_CONTEXT_LIMIT`; zero matched-budget violations.

Arm	Qwen3-8B (n=140)	Llama-3.1-8B (n=143)
FullKV	0.193	0.077
SnapKV	0.136	0.084
ADAPT (§7.3)	0.129	—
Structure	0.000	0.000
Uniform	0.000	0.000
Random	0.000	0.000

Is the harness sound? FullKV itself reaches only 0.193 and 0.077, so a reader should ask whether the pipeline is broken rather than the policy. Four checks argue otherwise. (i) Scoring is the unmodified upstream checker, and replaying the dataset's own ground truth through our rollout scores *valid*, while corrupted arguments and dropped turns score invalid. (ii) BFCL V3 multi-turn is known-hard for 8B models in prompting mode; low absolute accuracy is expected at this scale. (iii) The arms separate far above the floor: a Wilson upper bound of 0.027 on the zero arms excludes the FullKV rate on both models. (iv) An earlier configuration *did* return 0.000 for every arm including FullKV; we traced it to reasoning blocks reaching the decoder unstripped, fixed it, and FullKV rose to a non-degenerate rate. We report the diagnostic because it is the exact failure mode a skeptical reader should suspect.

Structure-aware retention scores zero on every conversation, on both models. Exact paired McNemar gives FullKV versus structure $p = 1.5 \times 10^{-8}$ (Qwen) and $p = 9.8 \times 10^{-4}$ (Llama). Attention selection, by contrast, is statistically indistinguishable from FullKV at a 4× budget: $p = 0.152$ on Qwen and $p = 1.0$ on Llama. Structure versus SnapKV is decisive against structure on both ($p = 3.8 \times 10^{-6}$ and $p = 4.9 \times 10^{-4}$). Two architectures reproduce every sign.

7.2 The protected fraction predicts the outcome

The failure has a measurable cause. A role-based policy can only *rank* while the protected mass is smaller than the keep budget; above that it selects everything and degenerates to index order. Figure 9 measures that quantity per workload with the same tagger used throughout.

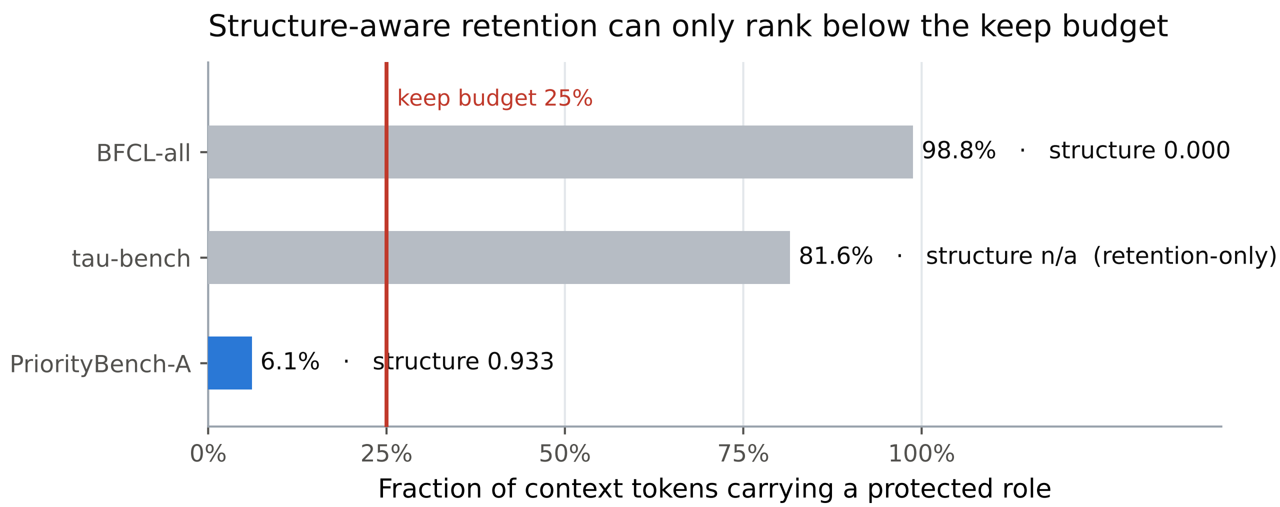

Figure 9: Protected mass relative to the keep budget. PriorityBench-A is 94.9% filler, so 6.1% of tokens carry a protected role and the policy has ample room to rank within a 25% budget. A BFCL system prompt *is* 32 JSON tool schemas, so 98.8% of tokens carry the protected TOOL role and nothing is available to discard. τ-bench sits between at 81.6%; it is a retention-only audit and has no task-success value.

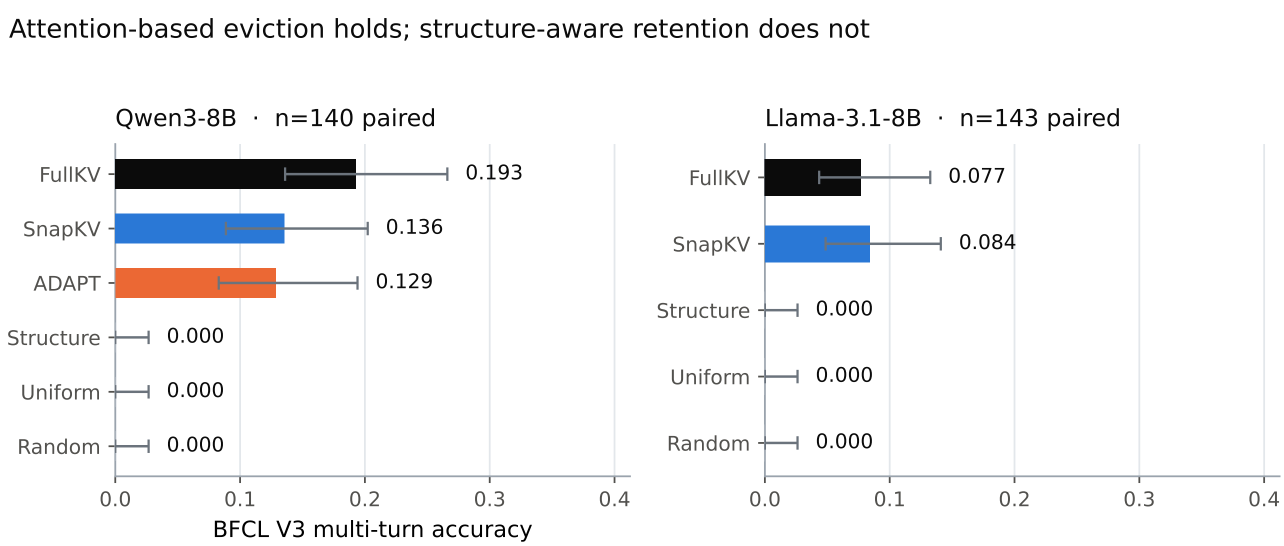

Figure 10: Paired BFCL accuracy with Wilson intervals. The same ordering holds on both models: FullKV and attention selection are indistinguishable, while the three non-attention policies are at zero with a Wilson upper bound of 0.027.

This bounds the PriorityBench-A result rather than contradicting it. The synthetic benchmark contains distractor filler by construction, which is exactly the regime in which "protect the structure, drop the filler" is a winning rule. Real tool-calling prompts contain no such filler.

A generation-free audit of 4,856 public τ-bench trajectories (828,000 extracted spans) corroborates the mechanism from the other side: structure retains explicit policy lines at 0.820 versus 0.001 for a position-only budget, but loses on reused identifiers (0.055 versus 0.315). Structure wins precisely where protected spans are scarce and separable.

7.3 ADAPT: structure as a budget-relative prior

The hybrid falsified in §6.6 force-protects structural positions, which consumes the entire budget whenever protected mass exceeds it. That is a mis-specification rather than a dead end. ADAPT replaces the hard constraint with a budget-relative prior,

$$\alpha = \min\left(1, \frac{\text{keep budget}}{\text{protected mass}}\right), \qquad s = \alpha \cdot \text{rank}(s_{\text{struct}}) + (1-\alpha) \cdot \text{rank}(s_{\text{attn}}), \tag{3}$$

where both quantities are known from the prompt, so α involves no fitting and no free parameter. Rank normalization is required because the structure and attention scores differ by several orders of magnitude. At $\alpha = 1$ the press provably selects exactly what the structure arm selects, so ADAPT subsumes the policy of §4 rather than approximating it.

On BFCL the measured α is 0.267 on average (range 0.250–0.401) over 833 generation steps, against ≈ 0.25 implied by the 98.8% protected fraction. ADAPT ties SnapKV (exact McNemar $p = 1.0$, $\Delta = -0.007$, 95% CI $[-0.057, +0.071]$) and is indistinguishable from FullKV ($p = 0.108$), while beating structure, uniform, and random ($p = 7.6 \times 10^{-6}$). We therefore claim only that ADAPT is never worse and recovers attention-level behavior automatically from a measurement; we do *not* claim it beats SnapKV. The α expression was written before the ADAPT run but committed afterwards, so the prediction is pre-specified by construction rather than by timestamp.

8 Discussion

Structure is a prior, not an oracle. The eviction results and retention audit jointly show that visible protocol structure can protect state that a sink/recent or random budget discards. The middle and buried controls give the necessary qualification: when relevant state remains recognizable, structure ties FullKV; when it is buried inside longer free-form turns, structure loses to FullKV. A production policy would need richer state metadata or a learned signal for unmarked spans.

The advantage depends on model and budget. Qwen at 25% keep strongly separates structure from position-only policies and places structure within four paired successes of SnapKV. Llama at 25% cannot differentiate any competent selector; at 5%, SnapKV is better on both slices. The tool-schema drop at that extreme budget suggests that role protection alone can allocate too much or too little within a broad protected span, while attention can choose a smaller subset. This is a useful crossover, not an anomaly to hide.

Hybrid complementarity is falsified at the tested settings. On Qwen, hybrid equals SnapKV; on Llama at 5%, it is below both parents. A union of protected positions with attention-ranked positions changes the residual budget and can inherit the weaknesses of both rules. Any future hybrid claim requires a new allocation rule and new evidence rather than reinterpretation of the current comparison. §7.3 supplies exactly that: the failure is that a hard union consumes the whole budget once protected mass exceeds it, and replacing the union with a budget-relative weight removes the pathology. We present ADAPT as a distinct rule tested on new data, not as a reinterpretation of the falsified hybrid.

The protected fraction is the operative variable. The external evaluation (§7) turns the model-and-budget dependence above into a single measurable quantity. Structure-aware retention helps when protected content is a minority of context and fails when it is the majority, because a binary role label cannot rank within a protected set that already exceeds the budget. The fraction is computable on CPU before any generation, which makes it a usable pre-flight check rather than a post-hoc explanation.

Bytes, peak, and speed are different systems quantities. Packed values and scale metadata genuinely reduce payload. Full-layer BF16 cold scratch then expands the live working set and adds preparation and decode cost. The observed 0.719× payload, 0.868× peak, and roughly 1.20× TPOT are mutually consistent. They do not establish serving throughput or capacity under concurrency.

9 Limitations and Threats to Validity

Synthetic workload. PriorityBench-A is generated from controlled templates and exact scorers. It measures whether particular protocol state survives, not the prevalence of these failures in deployed agent traffic, and it is not a LongBench or RULER matrix. §7 converts this from a caveat into a measurement: the benchmark's 94.9%

filler places it at one end of the protected-fraction axis, and the structure advantage does not survive at the other end.

External evaluation scope. The external results cover one budget (25% keep), two 8B models, and 150 conversations per model. BFCL V3 multi-turn is hard for models of this size, so FullKV itself scores 0.193 and 0.077; the arms are separated well above that floor, but absolute headroom is small. The τ-bench audit measures retention only—no user simulator, no tool backend, no generation—and excludes SnapKV, whose scores require realized attention.

Tagger–generator alignment. The tagger uses role, length, and schema/constraint markers that resemble the generator's visible templates. The gold audit rules out answer labels already in sink plus recent, but it does not eliminate all template alignment. The buried result—0.675, 0.650, and 0.675 versus FullKV 0.900, 0.875, and 0.900—is the clearest measured boundary.

9.1 Correction to a released position-blind baseline

Auditing our own released code against the paper surfaced one reporting error, which we correct here. The synthetic study listed "uniform" and "random" as two position-blind controls; they are one computation. In `select_random` the forced window is sized as recent = max(force_recent, budget − sink); whenever the budget exceeds sink + force_recent this makes the forced sink-plus-recent block exactly fill the budget, the residual is zero, and the sampling branch never executes. We verified byte-identical index sets at $n \in \{600, 4096, 40000\}$, with zero tokens ever drawn at random.

The consequence is contained. No measured value changes, because the column was always reporting the uniform policy correctly; what changes is the count of independent controls behind the position-blind comparison, from two to one. The structure-versus-position-blind separation—112/120 against 1/120—is unaffected in magnitude and significance. Our external evaluation uses a corrected position-blind arm that holds sink and recent at their configured sizes and samples the remainder; it also scores 0.000 on BFCL, so §7 stands as reported. We publish the correction because reproducibility claims are only worth the audits behind them.

Baseline fidelity. SnapKV, PyramidKV, H2O, and hybrid are matched-budget repository reimplementations, not runs of reference implementations. Chunked H2O under SDPA is particularly approximate. The paper therefore does not claim a reference-code leaderboard comparison.

Statistical scope. The paired Qwen structure–SnapKV result is not significant (p = 0.125). The eviction, baseline, and transfer studies are single canonical runs over deterministic example sets, not independent machine replications. Wilson intervals describe finite pass counts; they do not quantify hardware-run variability. The Llama 5% comparison has no tracked paired test.

Model and budget scope. Qwen3-8B on H200 is the primary matrix. Llama provides a ceiling and a negative crossover, not universal transfer. Gemma is a reduced 14-example legacy check with a low FullKV score. Only selected 5%, 25%, and legacy page budgets are tested.

Systems scope. The latency comparison uses different backends (FullKV SDPA and mixed FlashInfer) and single-request measurements. It does not measure batching, concurrent serving, throughput, or tails. Payload is not peak memory because cold attention expands INT4 into BF16. The streamed-cold variant records exit zero, parity, and a log peak, but it has no FullKV/TPOT frontier and is not a systems result.

10 Reproducibility

The repository freezes model revisions, manifests, configuration files, result bundles, and job commands. CPU logic and figures can be regenerated without an H200; GPU re-execution requires the pinned model weights and isolated GPU environment described in `docs/REPRODUCIBILITY.md`. Table 7 maps every headline claim to its tracked artifact.

The complete publication figure suite is generated from tracked artifacts by

```
uv run python scripts/make_publication_figures.py.
```

The script emits SVG and PDF files and renders full-size and 3.35-inch PNGs under `/tmp/prioritykv-figure-qc` for visual inspection. The paper compiles from either the repository root or `paper/`; the TeX source searches the appropriate figure directory in both layouts.

Table 7: Reproducibility index. Result paths are relative to `jobs/results/`; exact job commands appear in the final run manifest.

Item or claim	Pinned revision, digest, or result artifact
Qwen3-8B revision	`b968826d9c46dd6066d109eabc6255188de91218`
Llama-3.1-8B revision	`0e9e39f249a16976918f6564b8830bc894c89659`
Locked manifest (240)	SHA256 `fc44b966725738c94008ba61ce57ad7366169b9c0be73074f8161d909ccfae89`
Stress manifest (120)	SHA256 `154c07b900a7be6a26a0381b0fa28d49e0847086eade0b98e9472663b578784f`
Qwen eviction	Three `p0_w5_s*_kf25_token_*/summary.json` files
Placement controls	Six `p0[a-f]_w5_*_{middle,buried}_*/summary.json` files
Attention baselines	Three `p1_attn_baselines_s*_kf25_*/summary.json` files
H2O 82/120	`p1_h2o_chunked_s0_kf25_gpu1_r1/summary.json` plus baseline summaries for slices 1–2
McNemar $p = 0.125$	`p1_structure_vs_snapkv_mcnemar.json`
Gold-span audit	Six `audit_retention_*_kf25_summary.json` files
Llama transfer	`p3_llama31_attn_s*_kf{25,05}_*/summary.json`
Locked INT4 quality	`mg_b_lock240_quality_gpu01_r1/summary.json`
Payload and peak	`mg_a_peak_mem_gpu5_r1/summary.json`
Latency breakdown	`d4_latency_m3c_gpu56_r1/summary.json`
Streamed-cold smoke	`p2_fi_stream_cold_16k_gpu1_r1/log_full.txt`
Weak checks	`pub_c_gemma_reduced_gpu01_r6/summary.json`; `docs/atlas_w4_structure_rows.jsonl`

11 Conclusion

Application-visible structure is a strong retention prior *in a regime we can now identify in advance.* On synthetic agent traces it passes 112/120 at a 25% budget against 1/120 for position-blind eviction and is indistinguishable from SnapKV-class selection. On externally authored BFCL multi-turn traces the same policy scores 0.000 on two different 8B models, while attention selection remains indistinguishable from FullKV.

The reconciling variable is measurable without running the model: structure can rank only while protected content is a minority of the context. Our synthetic benchmark is 94.9% filler; real tool-calling prompts are almost entirely schema. We therefore recommend the protected fraction as a pre-flight check rather than structure-aware retention as a default policy, and we give a budget-relative rule, ADAPT, that degrades to attention selection exactly when the structural signal runs out.

The surrounding results delimit the claim rather than dilute it: structure matches but does not exceed FullKV under placement controls, role-aware INT4 placement is not a quality lever, and packed storage is an explicit space–time trade. Keeping these hypotheses separate is what makes the retention result interpretable. We also publish a correction to one released baseline, so every column here can be checked against the code. In operational terms the original intuition survives, narrowed: when a constrained cache must choose, retain the schema field before the pleasantry—but first measure whether the context is mostly schema, because then the rule has nothing left to discard.

References

[1] Ashish Vaswani et al. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017.

[2] Woosuk Kwon et al. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th ACM Symposium on Operating Systems Principles (SOSP)*, 2023. doi:10.1145/3600006.3613165.

[3] Guangxuan Xiao et al. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024. OpenReview:NG7sS51zVF.

[4] Zhenyu Zhang et al. H2O: Heavy-Hitter Oracle for efficient generative inference of large language models. *Advances in Neural Information Processing Systems*, 36, 2023.

[5] Yuhong Li et al. SnapKV: LLM knows what you are looking for before generation. *Advances in Neural Information Processing Systems*, 37, 2024. doi:10.52202/079017-0722.

[6] Zefan Cai et al. PyramidKV: Dynamic KV cache compression based on pyramidal information funneling. arXiv:2406.02069, 2024.

[7] Zirui Liu et al. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235:32332–32344, 2024.

[8] Zihao Ye et al. FlashInfer: Efficient and customizable attention engine for LLM inference serving. *Proceedings of Machine Learning and Systems*, 7, 2025.

[9] Yushi Bai et al. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 3119–3137, 2024. doi:10.18653/v1/2024.acl-long.172.

[10] Cheng-Ping Hsieh et al. RULER: What's the real context size of your long-context language models? In *First Conference on Language Modeling (COLM)*, 2024. OpenReview:kIoBbc76Sy.